MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON MIXER

TYPES: 1N23WE, 1N23WEM, 1N23WEMR,

1N23WG, 1N23WGM, 1N23WGMR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the detail requirements for a silicon point contact semiconductor diode for use as single units, matched forward pairs and matched forward and reverse pairs as a mixer in X band equipment.

* 1.2 Ratings and characteristics

	$\overline{F}$om		1/G	L	N	SWR
	1N23WE	1N23WG				
	dB	dB	ohms	dB	ratio	ratio
min	---	---	335	---	---	---
max	7.5	6.5	465	6.0	1.4	1.3

OPERATING AMBIENT TEMPERATURE: -65° to +150°C

STORAGE TEMPERATURE: -65°to +150°C

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

DRAWINGS

Defense Electronics Supply Center

D65019 - Test Holder for Microwave Diode Type 1N23.

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3. REQUIREMENTS

3.1 General. Requirements for semiconductor diodes shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500.

3.3 Design, construction, and dimensions. The semiconductor diode shall be of a symmetrical construction and shall consist of a cylindrical body having pins at both ends. The body outline with removable base adapter shall conform to figure 1.

3.3.1 Base adapter. The base adapter shall be of the design, construction, and physical dimensions shown on figure 2.

3.3.2 Plating. The diode shall be plated as specified on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

3.5.1 Matched diodes. The "M" suffix for matched diodes shall be omitted in the type designation on each device. Matched diodes meeting the requirements of this specification shall be packaged with a statement to that effect.

* 3.6 Burnout by multiple RF pulses. At the end of manufacturing processes and prior to selecting samples for testing, all diodes shall be subjected to 100 percent burnout by exposure to a series of RF pulses with the test conditions as follows:

P = 20 watts
f = 8.8 to 9.8 GHz
R_L = 100 ohms
t_p = 6 ±3 nanoseconds
Exposure = 15,000 pulses min
Holder: See below

Test data demonstrating that the holder used meets this requirement shall be made available to the government representative upon request (see figures 3 and 4).

After the burnout by multiple pulse, group A, subgroup 2 tests shall be performed on all devices in the lot. Any device which fails any of these tests shall be eliminated from the lot.

4. QUALITY ASSURANCE PROVISIONS

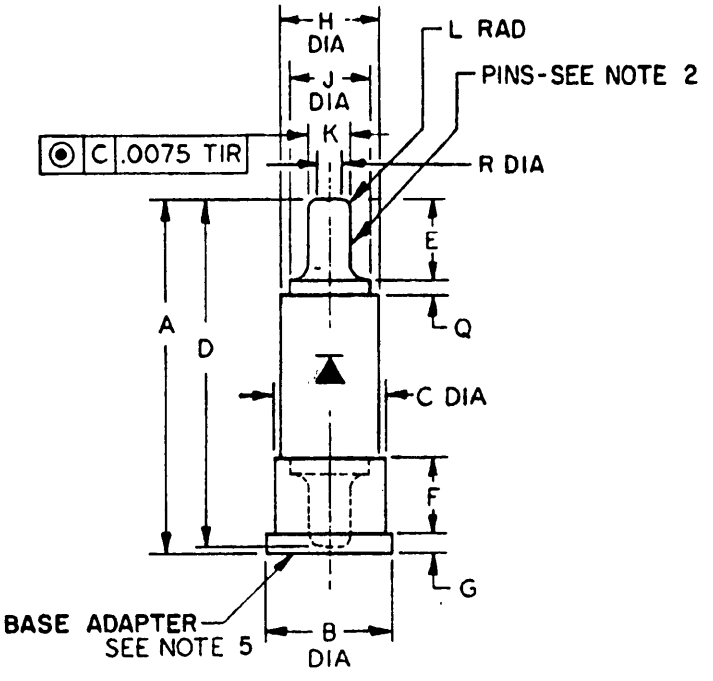
4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

* 4.1.1 Structurally similar device types. Types 1N23WE and 1N23WG are structurally similar devices. Any type may be sampled for groups B and C inspections to qualify and accept all types for qualification inspection and quality conformance inspection.

* 4.2 Test conditions. Unless otherwise specified herein, the test conditions, when applicable, shall be as follows:

P = 1.0 mW ±5%
$N_{if} \geq$ 1.5 dB
R_L = 100 ± 1 Ω
f = 9375 ± 5 MHz
Z_m = 400 ± 10Ω; ± j0Ω
Drawing D65019

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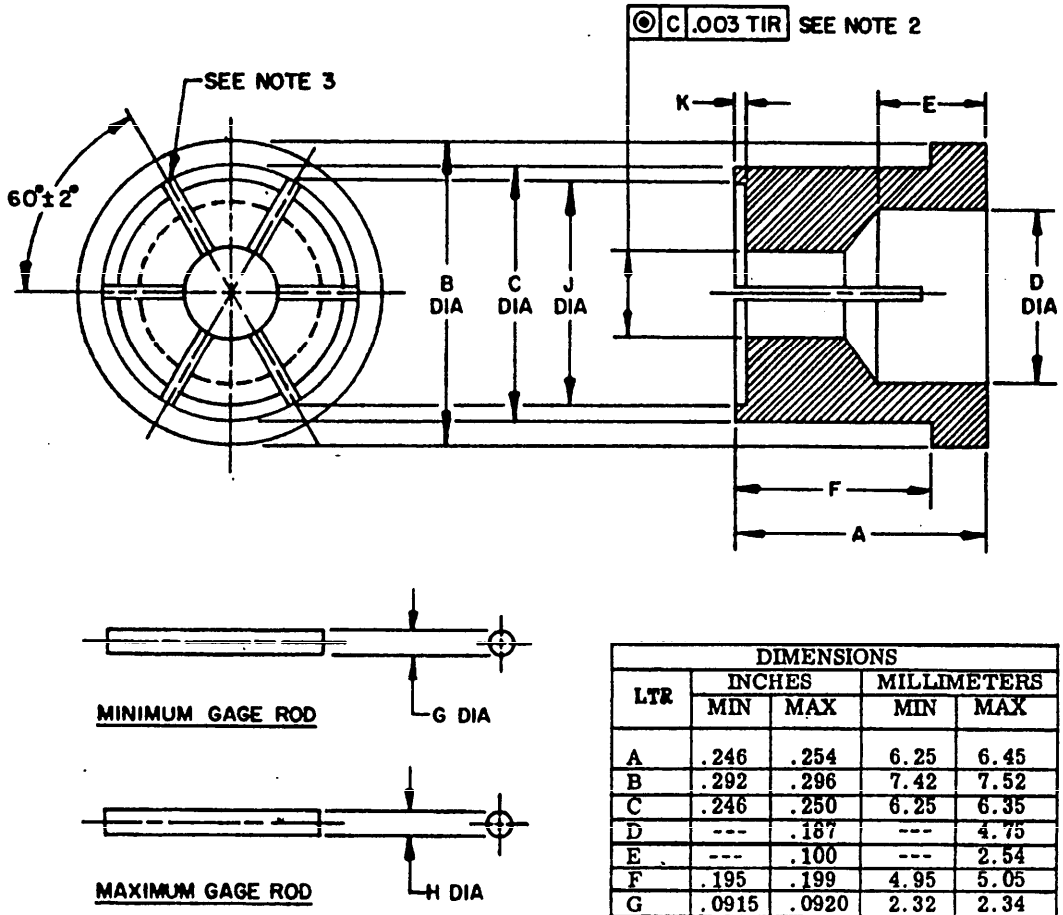


DIMENSIONS					
LTR	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A	.800	.840	20.32	21.34	
B	.292	.296	7.42	7.52	
C	.246	.250	6.25	6.35	
D	.766	.792	19.46	20.12	
E	.180	.190	4.57	4.83	6
F	.193	.199	4.90	5.05	
G	.047	.057	1.19	1.45	
H	.222	.240	5.64	6.10	
J	.195	.215	4.95	5.46	6
K	.092	.094	2.34	2.39	2, 5, 6
L	.030	.046	.76	1.17	6
Q	.020	.030	.51	.76	6
R	---	.030	---	.76	6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. The ends of both pins shall be smooth with no burrs or sharp edges. Within the diameter R a recess is allowed on the ends of the pins.
3. Metal parts shall be gold plated 10 msi minimum.
4. The polarity shall be indicated by an arrow, with the arrow pointing in the direction of easier current flow.
5. Removable base adapter shall establish the forward or reverse polarity.
6. Applies to both pins.

FIGURE 1. Semiconductor device, diode, types 1N23WE, 1N23WEM, 1N23WEMR, 1N23WG, 1N23WGM, and 1N23WGMR.

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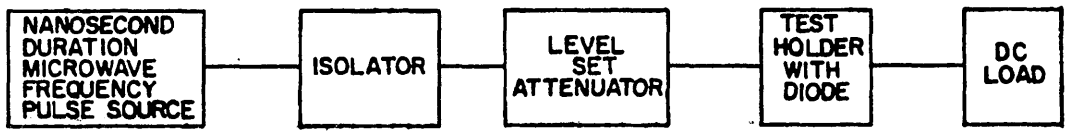


LTR	DIMENSIONS			
	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.246	.254	6.25	6.45
B	.292	.296	7.42	7.52
C	.246	.250	6.25	6.35
D	---	.187	---	4.75
E	---	.100	---	2.54
F	.195	.199	4.95	5.05
G	.0915	.0920	2.32	2.34
H	.0940	.0945	2.39	2.40
J	.216	.221	5.49	5.61
K	.031	.036	.79	.91

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. This diameter should be .100 (2.54 mm) max diameter before closing jaws. This diameter shall then be sufficiently closed, and the adapter so tempered that it will fit on the minimum gage rod and also on the maximum gage rod, in each case with a snug fit (for hand assembly).
3. Each slot shall be .013 (.330 mm) min, .017 (.432 mm) max wide by .185 (4.70 mm) deep before closing. The 6 slots are equally spaced.
4. The material for the adapter shall be beryllium copper or half-hard brass per QQ-B-626, composition 22, and shall be gold plated 10 msi minimum.
5. All burrs and sharp edges shall be removed.

FIGURE 2. Removable base adapter, included as a part of types 1N23WE, 1N23WEM, 1N23WEMR, 1N23WG, 1N23WGM, and 1N23WGMR.

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NOTES:
1. The holder used with a diode mounted in it and terminated with 100 ohms and with an incident power of 1.0 mW to the holder shall have a 2.0 maximum VSWR over a minimum band width of $\frac{1}{\text{pulse width used}}$ on both sides of the RF burnout test frequency used.
2. A suitable circuit for generating nanosecond pulses of the required power is described in the 1969 IEEE-G-MTT International Microwave Symposium Conference Record. (See figure 4).

FIGURE 3. Multiple RF pulse burnout circuit.

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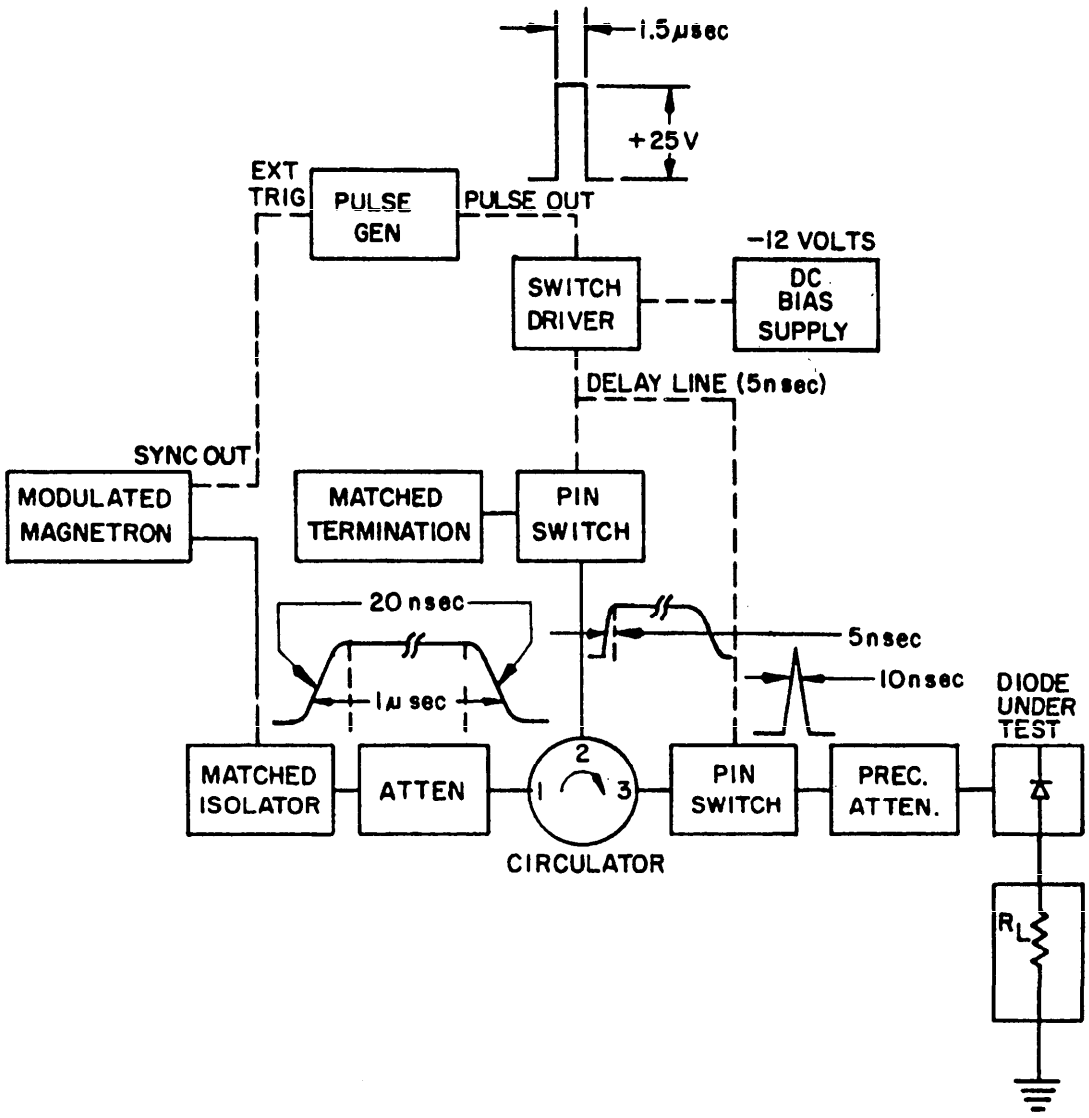


FIGURE 4. RF pulse burnout system.

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4.2.1 Holder. The diode holder specified in Drawing D65019, or equivalent, shall be used for all electrical test measurements.

* 4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III, except the matched-pair requirements in subgroup 3, table I.

* 4.3.1 Qualification testing. If the qualification inspection lot is made up of sublots of 1N23WE and 1N23WG, each subgroup of groups B and C shall be made up of an equal (or near equal as possible) number of devices from each sublot. The manufacturer may, at his option, use the same sample for more than one subgroup of the group B and group C tests. All units must meet the group A test limits prior to submission to a new subgroup in the group B tests. The manufacturer need not make more units for the qualification tests than the total number of devices he plans to submit to groups A, B, and C testing.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in groups A, B, and C.

4.4.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.4.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.4.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in tables, I, II, and III, and as follows:

* 4.5.1 High-temperature operation. The semiconductor diode shall be placed in the mixer holder. The ambient temperature of the diode, with test conditions specified for the overall noise figure, shall be raised to +150°C and maintained at this temperature until equilibrium is reached. The $\overline{F}_{om}$ shall then be determined and shall be less than the specified limit. The temperature shall then be returned to 25 ± 3°C at which time $\overline{F}_{om}$ shall be less than the specified limit.

* 4.5.2 Matched pair, "M" suffix (forward polarity). The matched pair (M suffix) shall consist of two diodes, tested to requirements of subgroup 3, table I, having the anode connected to the adapter.

* 4.5.3 Matched pair, "MR" suffix (forward and reverse polarity). The matched forward and reverse pair (MR suffix) shall consist of two diodes, tested to requirements of subgroup 3, table I: one diode having the anode connected to the adapter (forward polarity) and the second diode having the cathode connected to the adapter (reverse polarity).

4.5.4 Time limit for end points. End point tests for qualification and quality conformance inspection shall be completed within 96 hours after completion of the last test in the subgroup.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			7				
Visual and mechanical examination	2071			---	---	---	---
* Subgroup 2			5				
* Voltage standing wave ratio (see 4.2)	4136	I = 1.0 mAdc		VSWR	---	1.3	ratio
* IF conductance (see 4.2)	4116	f = 60 - 2,000 Hz		1/G	335	465	ohms
* Overall noise figure (see 4.2)	4126	Test condition A		$\overline{F}_{om}$			
1N23WE					---	7.5	dB
1N23WG					---	6.5	dB
Subgroup 3			5				
Matched pair requirements: (see 4.5.2 and 4.5.3)							
Conversion loss unbalance	4101			ΔL	---	0.3	dB
Intermediate frequency conductance unbalance	4116			$\Delta 1/G$	---	25	ohms

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TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1							
Physical dimensions	2066	(See figure 1 and 2)					
		Dim. C, J, and K of figure 1	7	---	---	---	---
		Dim. A, B, D, E, F, G, H, L, and Q of figure 1 (Dim. N, and M and note 3 of figure 1 and note 4 of figure 2 are for qualification only).	20	---	---	---	---
Subgroup 2			15				
Thermal shock (temperature cycling)	1051	Test condition F		---	---	---	---
Terminal strength:							
Tension	2036	Test condition A; 1 lb; t = 30 sec		---	---	---	---
Torque	2036	Test condition D_1; t = 30 sec		---	---	---	---
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
* End points: (see 4.5.5)							
Overall (average) noise figure (see 4.2)	4126	Test condition A		$\overline{F}_{om}$	---	---	---
1N23WE						8.5	dB
1N23WG						7.5	dB
* Subgroup 3			15				
* Shock	2016	Nonoperating; 500 G; t = 1.0 msec; 5 blows in each orientation: X_1, Y_1, and Y_2		---	---	---	---
* Vibration, variable frequency	2056	Nonoperating; 15 G, 50 to 2,000 Hz		---	---	---	---
End points: (Same as subgroup 2)				---	---	---	---

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TABLE II. Group B inspection. -Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 4			10				
High temperature operation (see 4.5.1)	---	T_A = 150°C		---	---	---	---
Overall (average) noise figure		Test Condition A		$\overline{F}_{om}$			
1N23WE						10.5	dB
1N23WG						9.5	dB
Overall (average) noise figure		Test condition A T = 25°C		$\overline{F}_{om}$			
1N23WE						8.5	dB
1N23WG						7.5	dB
* Subgroup 5							
High temperature life (nonoperating)	1031	T_A = 150°C		---	---	---	---
End points: (Same as subgroup 2)							

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TABLE III. Group C inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Conversion loss (see 4.2)	4101	Dwg. D65019		L	---	6.0	dB
Output noise ratio (see 4.2)	4121	Dwg. D65019; R_e = 300 ohms		N	---	1.4	ratio

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6.0 NOTES

6.1 The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Superseded diode types. Superseded diode types 1N3747 shall be replaced by the 1N23WE.

* 6.3 Changes from the previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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